

Goal: Construct X s.t. $\text{RIP}(X, s, 1/3) \leadsto \frac{T}{F}$

in a Probabilistic $\forall \delta > 0$. $\underset{n \gg N_0}{P}_X(\text{RIP}(X, s, 1/3)) \geq 1 - \delta$.

$$n \approx O(s \log d/s)$$

$X_{ij} \stackrel{\text{iid}}{\sim} \text{Rad.}$

$$P_X(\text{MIC}(X, 1/3s)) \geq 1 - \delta \quad n \approx O(s^2 \log s)$$

$$\text{RIP}(X, s, 1/3) : \forall s: |s| \leq s \quad \left\| \underbrace{\frac{1}{n} X_s^T X_s - I}_A \right\|_2 \leq 1/3.$$

$$X_{n \times d} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \begin{bmatrix} \underbrace{\hspace{10em}}_{d \gg n} \end{bmatrix}$$

Subgoal: Show that $\|A\|_2$ Concentrates around 0.

$$P\left(\|A\|_2 \leq c \sqrt{\frac{s + \log 1/\delta}{n}}\right) \geq 1 - \delta \quad \forall \delta > 0.$$

$$\rightarrow \|A\|_2 = \sup_{v \in S^{s-1}} |v^T A v|$$



$$\rightarrow v^T A v = v^T \left(\frac{X^T X}{n} - I \right) v = \frac{v^T X^T X v}{n} - \|v\|^2$$

$$= \frac{1}{n} \langle Xv, Xv \rangle - \|v\|^2$$

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \left(\begin{array}{l} \text{from now on} \\ X \text{ is the same} \\ \text{as } X_S \end{array} \right)$$

$$Xv = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} v \\ \vdots \\ v \end{bmatrix} = \begin{bmatrix} x_1^T v \\ \vdots \\ x_n^T v \end{bmatrix}$$

$$\Rightarrow v^T A v = \frac{1}{n} \sum_{i=1}^n (x_i^T v)^2 - \|v\|^2$$

$$(x_i^T v)^2 \quad i=1 \dots n \quad \text{iid copy}$$

$$x_i^T v \quad \text{sub-gaussian R.V.}$$

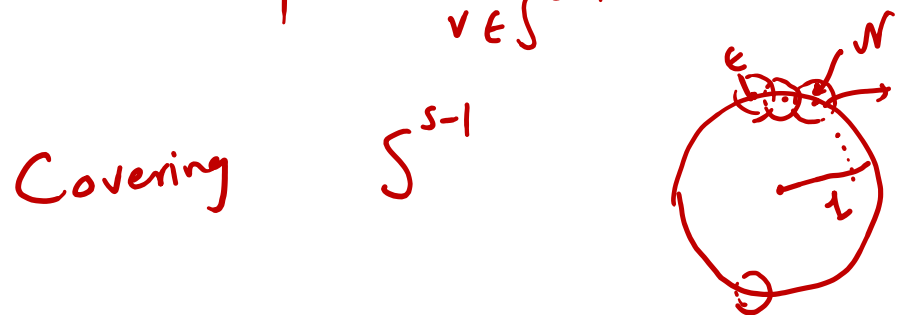
$$x_{ij} \sim \text{Sub G}(\sigma=1)$$

$$x_i^T v = \sum_j v_j x_{ij} \rightarrow \text{sub G}(\sigma = \sqrt{|v_j|})$$

$$\sim \text{sub G}(\sigma = \sqrt{\sum_j v_j^2})$$

Eq. (2.11) in Wainwright $\forall t > 0: P(|v^T A v| > t) \leq 2 e^{-c n t^2}$ $P_1 \Rightarrow Q_1$

$$P\left(\sup_{v \in S^{s-1}} |v^T A v| > t\right) \leq P\left(\exists v \in \underbrace{S^{s-1}}_{Q_1} |v^T A v| > t\right)$$



$$N = \{x \in S^{s-1}\} \quad |N| \leq \left(\frac{3}{\epsilon}\right)^s$$

$$P\left(\exists v \in N(S^{s-1}, \epsilon) : |v^T A v| > t\right) \leq 2 \left(\frac{3}{\epsilon}\right)^s e^{-c n t^2}$$

$u \notin N$ $\hookrightarrow u \in S^{s-1}$ $\|v - u\| \leq \epsilon$ (pick $v \in N$)

$$\begin{aligned} |u^T A u| &= |v^T A v - v^T A v + u^T A u| \leq |v^T A v| + |(u-v)^T A u + (u-v)^T A v| \\ &\leq |v^T A v| + 2\epsilon \|A\|_2 \leq t + 2\epsilon \|A\| \end{aligned}$$

$$\begin{array}{l} \overbrace{\forall v \in \mathcal{N}}^P \quad \checkmark \\ |v^T A v| < t \\ \uparrow \end{array} \Rightarrow$$

$$\|A\|_2 \leq t + 2\epsilon \|A\|_2 \Rightarrow \|A\|_2 \leq \frac{t}{1-2\epsilon}$$

$$\epsilon = 1/4 \rightarrow \overbrace{\|A\|_2 \leq 2t}^Q \quad \checkmark$$

$$\begin{array}{l} P \Rightarrow Q \\ \neg Q \Rightarrow \neg P \end{array}$$

$$\mathbb{P}(P) \leq \mathbb{P}(Q)$$

$$\mathbb{P}(\neg Q) \leq \mathbb{P}(\neg P)$$

$$\underbrace{\mathbb{P}(\|A\|_2 > 2t)} \leq \mathbb{P}(\exists v \in \mathcal{N} |v^T A v| > t) \leq \underbrace{2 \cdot 12^s e^{-cnt^2}}$$

$$A = \frac{X_s^T X_s}{n} - I$$

$$\underbrace{\mathbb{P}(\neg \text{RIP}(X, s, t))}_\delta \leq \underbrace{2 \cdot s \binom{d}{s} \cdot 12^s e^{-cnt^2}}$$

$$\binom{d}{s} \approx \left(\frac{d}{s}\right)^s \quad s \ll d$$

$$\delta = 2 \cdot s \cdot \left(\frac{d}{s}\right)^s \cdot 12^s e^{-cnt^2} \quad t = 1/6$$

$$\log 1/\delta = -\log 2s - s \log d/s - s \log 12 + cnt^2$$

$$n \sim O\left(s \log d/s + \log(1/\delta)\right)$$

□

$$|N| \leq \left(\frac{3}{\epsilon}\right)^s.$$



$N =$ maximal set of $z \in \mathbb{S}^{s-1}$

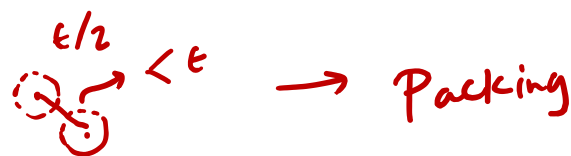
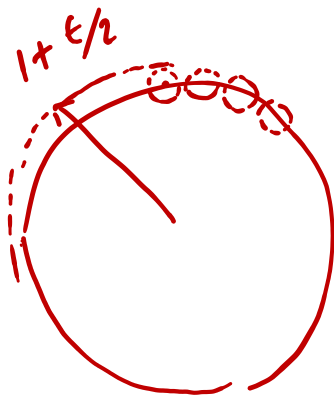


s.t. $z_1, z_2 \in N$

$$\|z_1 - z_2\| > \epsilon$$

$\forall z \in S^{s-1}, \exists x \in \mathcal{N} : \|x - z\| \leq \epsilon \Rightarrow \mathcal{N}$ is also
 $\subset$

a covering.



$$\bigcup_{z \in \mathcal{N}} B(z, \epsilon/2) \subseteq \underbrace{B(0, 1 + \epsilon/2)}$$

$$|\mathcal{N}| \cdot \left(\frac{\epsilon}{2}\right)^s \text{vol}_s(1) \leq (1 + \epsilon/2)^s \text{vol}_s(1)$$

$$|\mathcal{N}| \leq \left(\frac{2}{\epsilon} + 1\right)^s \leq \left(\frac{3}{\epsilon}\right)^s$$